

Elizabeth Yankovsky

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EDUCATION

Princeton University, Princeton, NJ

Ph.D., Atmospheric and Oceanic Sciences, 2015-2020

Thesis: Modeling & parameterizing submesoscale turbulence in dense Arctic flows

Advisor: Dr. Sonya Legg

University of South Carolina Honors College, Columbia, SC

B.S., Physics and Geophysics, 2011-2015

Thesis: Methane hydrates and cellular convection in the Central Aleutian Basin

Advisors: Drs. Camelia Knapp and Darrell Terry

APPOINTMENTS

Yale University, New Haven, CT

Assistant Professor in Earth and Planetary Sciences, July 2024-present

[C]Worthy, Boulder, CO

Postdoctoral Researcher, 2023-2024

Mentors: Drs. Matthew Long, Alicia Karspeck, Scott Bachman

Courant Institute, New York University, New York, NY

Postdoctoral Associate, Center for Atmosphere Ocean Science, 2020-2023

Mentors: Drs. Laure Zanna and Shafer Smith

NOAA Geophysical Fluid Dynamics Laboratory, Princeton University

Graduate Research Assistant, Ocean & Ice Processes Group, 2015-2020

Committee: Drs. Sonya Legg, Robert Hallberg, Stephen Griffies, Rong Zhang

Geophysical Exploration Laboratory, University of South Carolina

Undergraduate Research Assistant, 2012-2015

Mentors: Drs. Camelia Knapp, Darrell Terry

Oregon State University, College of Earth, Ocean, & Atmospheric Sciences

NSF-REU program intern, June-August 2014

“Response of the Length and Stratification of the North River Estuary to Changes in Forcing”, Mentor: Dr. James Lerczak

Rutgers University, Department of Marine and Coastal Sciences

NSF-REU program intern, June-August 2013

“Quantifying Turbulent Dissipation in a Shallow Estuarine Environment”

Mentor: Dr. Robert Chant

AWARDS

2025 AGU Outstanding Reviewer Award for JAMES

2024 Council of the American Meteorological Society Editor’s Award – JPO

2017 National Science Foundation Graduate Research Fellowship

2011 National Merit Scholarship

TEACHING

Yale EPS 5360: Waves, Instabilities, and Turbulence. Fall 2025.

Yale EPS 2160: Global Warming: Climate Physics. Co-taught with Dr. John Wettlaufer, Spring 2025, Spring 2026.

Princeton Instructor Assistant: Introduction to Ocean Physics for Climate (GEO-MAE 425). Taught by Dr. Gabriel Vecchi, Fall 2018.

Teaching Transcript Program, Princeton McGraw Center, completed 2020.

STUDENT AND POSTDOCTORAL SUPERVISION	Postdoctoral Advisees: Dr. Jago Strong-Wright (2026-present) Dr. Sadegh Ranjbar (2026-present) Dr. Qi Zhang (2026-present) Dr. Hassan Mason (2026-present) Graduate Advisees as Primary Advisor: Yiming Bian (Ph.D. expected 2029) Suhani Dalal (Ph.D. expected 2029) Krithin Gowthaman (Ph.D. expected 2030) Anastasiia Chupakhina (M.S., 2026) Graduate Advisees as Minor Advisor: Jennifer Kosty (Ph.D. expected 2028) Yanni Bills (Ph.D. expected 2029) Maggie Yao (Ph.D. expected 2029) Undergraduate Advisees: Jonathan Cohen (Yale College '27) Lucy O'Brien (Yale College '29) Thesis Committee Membership (Advisees Excluded): Nico Theunissen, Sam De Abreu, Demetra Yancopoulos, Jelena Collins
FUNDING HISTORY	Carbon to Sea Model Skill Assessment Workshop Mini-grant, 2026. Quantifying Uncertainty from Atmospheric Forcing Resolution in Regional Ocean Alkalinity Enhancement Models. PI: Elizabeth Yankovsky. Bezos Earth Fund: AI for Climate and Nature Grand Challenge, 2026-28. Leveraging AI to Accurately and Rapidly Predict Responses to Open-System Geochemical Climate Interventions. Phase I and Phase II awardee. Lead PI: Elizabeth Yankovsky; additional PIs: Luke Gloege, Noah Planavsky. Yale Planetary Solutions, 2026-27. Oceanic Carbon Capture using Hydrogen Looping: Bridging Electrochemical Engineering with Earth System Modeling. Lead PI: David Kwabi. Yale Planetary Solutions, 2025-26. Validation of a Novel Solid State Alkalinity Sensor for Marine Carbon Dioxide Removal. Lead PI: Peter Raymond. Yale Planetary Solutions, 2025-26. Climate-Driven Global Expansion of Dengue Endemicity. Lead PI: Nathan Grubaugh.
PUBLICATIONS	Yankovsky, E., 2026: Near-surface mesoscale and submesoscale ocean dynamics. <i>Encyclopedia of Climate Science, Chapter 2.1.6</i> [in press]. Zhou, M., M. Timmermans, E. Yankovsky, M. Tyka, and M. Long, 2026: Global intercomparison of ocean-based geochemical carbon dioxide removal efficiencies. <i>Nature Communications</i> [in press]. Tyka, M., M. Zhou, E. Yankovsky, and D. Carroll, 2026: Substantial inter-model variation in OAE efficiency between the CESM2/MARBL and ECCO-Darwin ocean biogeochemistry models. <i>Biogeosciences</i>. Steinberg, J., E. Yankovsky, S. Cole, L. Zanna, 2025: A landscape of mesoscale eddy vertical structure: the influence of bathymetric slope and roughness on kinetic energy. <i>Journal of Physical Oceanography</i>.

Yankovsky, E., M. Zhou, M. Tyka, S. Bachman, D. Ho, A. Karspeck, and M. Long, 2025: Impulse response functions as a framework for quantifying ocean-based carbon dioxide removal. *Biogeosciences*.

Zhou, M., M. Tyka, D. Ho, **E. Yankovsky**, S. Bachman, T. Nicholas, A. Karspeck, M. Long, 2024: Mapping the global variation in the efficiency of ocean alkalinity enhancement for carbon dioxide removal. *Nature Climate Change*.

Yankovsky, E., S. Bachman, K. S. Smith, L. Zanna, 2024: Vertical structure and energetic constraints for a backscatter parameterization of ocean mesoscale eddies. *Journal of Advances in Modeling Earth Systems*.

Yankovsky, E., A. Yankovsky, 2023: The cross-shelf regime of a wind-driven supercritical river plume. *Journal of Physical Oceanography*.

Yankovsky, E., L. Zanna, K. S. Smith, 2022: Influences of mesoscale ocean eddies on flow vertical structure in a resolution-based model hierarchy. *Journal of Advances in Modeling Earth Systems*.

Marques, G., N. Loose, **E. Yankovsky**, J. Steinberg, C. Chang, N. Bhamidipati, A. Adcroft, B. Fox-Kemper, S. Griffies, R. Hallberg, M. Jansen, H. Khatri, L. Zanna, 2022: NeverWorld2: An idealized model hierarchy to investigate ocean mesoscale eddies across resolutions. *Geoscientific Model Development*.

N. Loose, R. Abernathey, I. Grooms, J. Busecke, A. Guillaumin, **E. Yankovsky**, G. Marques, J. Steinberg, A. S. Ross, H. Khatri, S. Bachman, L. Zanna, P. Martin, 2022: GCM-Filters: A Python package for diffusion-based spatial filtering of gridded data. *Journal of Open Source Software*.

I. Grooms, N. Loose, R. Abernathey, J. Steinberg, S. Bachman, G. Marques, A. Guillaumin, **E. Yankovsky**, 2021: Diffusion-based smoothers for spatial filtering of gridded geophysical data. *Journal of Advances in Modeling Earth Systems*.

Yankovsky, E., S. Legg, R. Hallberg, 2021: Parameterizing submesoscale symmetric instability and frontal mixing in dense flows along topography. *Journal of Advances in Modeling Earth Systems*.

Yankovsky, E., S. Legg, 2019: Symmetric and baroclinic instability in dense shelf overflows. *Journal of Physical Oceanography*.

Yankovsky, E. A., D. A. Terry, C. C. Knapp, 2015: Seismic and gravity evidence for methane-hydrate systems in the central Aleutian Basin. *International Journal of Earth Science and Geophysics*.

SELECTED
SEMINARS AND
TALKS

INVITED TALKS

Modeling the Mediterranean Overflow Water Pathways: *International Ocean Discovery Program Expedition 401 Post-Cruise Meeting*, Granada, Spain, 2026.

Physical Oceanography Perspectives on Marine Carbon Dioxide Removal: *Physical Oceanography Spring Seminar*, University of Rhode Island, 2026 and *Department of Atmospheric and Oceanic Sciences Colloquium Series*, University of Wisconsin-Madison, 2025.

Bridging scales in the ocean carbon cycle: *Ocean Carbon and Biogeochemistry 2025 Summer Science Workshop*, Woods Hole Oceanographic Institute, 2025.

Challenges & applications of ocean modeling: *BIOMES Seminar Series*, Yale School of the Environment, 2025.

Geophysical fluid dynamics perspectives on marine carbon dioxide removal: *Physical Oceanography Seminar*, Woods Hole Oceanographic Institution; *Physics Colloquium*, Brandeis University, 2025.

Links between eddy horizontal and vertical structure: a geostrophic turbulence interpretation. *Climate Applications of Layering Workshop*, Isaac Newton Institute for Mathematical Sciences, University of Cambridge; and *Theoretical and Practical Perspectives in Geophysical Fluid Dynamics Program*, International Center for Theoretical Sciences, Bengaluru, India (virtual), 2024.

Impulse response functions as a framework for quantifying carbon uptake associated with ocean alkalinity enhancement. *Spring Symposium on Marine Carbon Dioxide Removal*, Yale University (YCNCC), 2024.

A backscatter-only parameterization for mesoscale eddies. *IUGG 2023 General Assembly*, Berlin, Germany, 2023.

The vertical structure of mesoscale ocean eddies. *Atmosphere Ocean Science Colloquium*, Courant Institute, New York University, 2023.

Improving ocean models across scales: techniques, progress, and open questions. *AOCD Spring Seminar Series*, Yale University, 2023.

The role of ocean turbulence in climate. *The Department of Earth and Planetary Sciences Colloquium*, Yale University, 2023.

Modeling & parameterizing mesoscale eddies in the Arctic Ocean. *US Interagency Arctic Research Policy Committee* modeling team meeting, 2022.

Parameterizing mesoscale eddy energetics and vertical structure at eddy-permitting resolutions. *NCAR Oceanography Seminar*, Boulder, CO, 2022.

Influences of mesoscale ocean eddies on flow vertical structure. *Oceans Research Group Seminar*, University of Oxford, 2022.

Modeling and parameterizing submesoscale turbulence in dense Arctic overflows. *Atmosphere, Ocean and Climate Sack Lunch Seminar Series*, MIT EAPS, 2021.

Constraining water mass transformation and overflow dynamics on the Arctic shelves. *Polar Oceans Seminar Series*, British Antarctic Survey, 2021.

Symmetric instability in Arctic dense gravity currents. *Seminar in Applied and Computational Mathematics*, University of Edinburgh, Scotland, 2019.

Modeling baroclinic and submesoscale instabilities in the Arctic Ocean. *AOCD Fall Seminar Series*, Yale University, 2018.

CONFERENCE PRESENTATIONS

Multi-resolution regional modeling to investigate variability in the carbon uptake efficiency of ocean alkalinity enhancement and implications for prediction. *AGU Fall Meeting, Ocean Sciences Meeting 2025/2026*.

Mesoscale turbulence in the context of marine carbon dioxide removal. *Climate Process Team Annual Meeting: Ocean Transport and Eddy Energy*, Brown University, 2024.

Evaluation of “Impulse Response Functions” as a framework for quantifying carbon uptake associated with ocean alkalinity enhancement. *Ocean Sciences Meeting*,

New Orleans, LA, 2024 and *Ocean Model Working Group Meeting*, NCAR, Boulder, CO, 2024.

Exploring mesoscale eddy vertical structure regimes in the global ocean. *AGU Fall Meeting*, 2022.

Influences of mesoscale ocean eddies on flow vertical structure. *Ocean Sciences Meeting*, 2022; *Climate Process Team Annual Meeting: Ocean Transport and Eddy Energy*, Boulder, CO, 2022.

Effects of eddy representation on vertical structure and energetics. *CESM Ocean Model Working Group Meeting*, 2021.

Constraining Arctic water mass transformation and ventilation pathways in the GFDL-OM4.0. *AGU Fall Meeting*, 2020.

Modeling vertical transport and submesoscale frontal mixing in dense flows along topography. *Ocean Sciences Meeting*, San Diego, CA, 2020.

Symmetric and baroclinic instability in dense shelf overflows. *EGU General Assembly*, Vienna, Austria, 2019.

Symmetric instability in dense shelf overflows. *Ocean Sciences Meeting*, Portland, OR, 2018.

Dense water formation and transport on the Arctic continental shelves. *Forum for Arctic Ocean Modeling and Observational Synthesis (FAMOS)*, Woods Hole Oceanographic Institution, MA, 2017.

Response of the length and stratification of the North River estuary to changes in forcing. *AGU Fall Meeting*, San Francisco, CA, 2014.

PROFESSIONAL
SERVICE

Associate Editor for Journal of Physical Oceanography: 2025-present.

Associate Editor for Journal of Geophysical Research: Machine Learning and Computation, 2024-present.

Reviewer for: Journal of Physical Oceanography, Ocean Modelling, Geoscientific Model Development, Journal of Advances in Modeling Earth Systems, Geophysical Research Letters, Environmental Fluid Mechanics, Earth's Future, Journal of Geophysical Research, Nature Communications, NSF, Fram Centre Research Cooperation, NASA NSPIRES program.

Convener and chair for the session “Multiscale Eddy Dynamics and Tracer Transport: Bridging Observations, Theory, and Modeling” at Ocean Sciences Meeting, 2024.

Convener and chair for the session “Mesoscale Eddy Energy and Ocean Transport” at Ocean Sciences Meeting, 2022.

UNIVERSITY
SERVICE

University Corporation for Atmospheric Research (UCAR): Yale University member representative, 2024-present.

Joint Yale Committee for the Marshall and Rhodes Scholarships, Fall 2025.

Scientific Leadership Team: Yale Center for Natural Carbon Capture.

Faculty Affiliate: Yale Institute for Biospheric Studies.

Co-organizer: Yale Atmosphere, Oceans, Climate Dynamics seminar, 2024-26.

NYU-CAOS Colloquium Organizing Committee (2022), **Planning Committee** for the yearly Princeton AOS Program Orientation and Retreat (2018), AOS Program **Student-Faculty Representative** (2017-18).

WORKSHOPS

Carbon to Sea Model Skill Assessment Workshop, Steering Committee member. Garmisch-Partenkirchen, Germany, June-July 2026.

AI for Natural Climate Solutions, Organizer. Yale Center for Natural Carbon Capture & Yale Institute for Biospheric Studies, May 2025.

Climate Applications of Layering, Invited Participant. Isaac Newton Institute for Mathematical Sciences, University of Cambridge, May 2024.

Machine Learning and Climate Modeling: Princeton AOS, July 2019.

Convection in Nature: Princeton Center for Theoretical Science, Feb. 2018.

Forum for Arctic Modeling and Observational Synthesis (FAMOS): Woods Hole Oceanographic Institution, Oct. 2017.

Les Houches Summer School on Fundamental Aspects of Turbulent Flows in Climate Dynamics: Les Houches Physics School, Aug. 2017.

OUTREACH

Yale-New Haven Teachers Institute: developing lectures to present to local public school teachers, 2024-present. Gave lecture titled: “How Supercomputing & Data Availability Are Changing How We Understand Our Planet”, 2025.

Hugo Mentorship Program: mentoring high school students on physical oceanography and climate science summer projects, 2025-present.

STEM Professionals Day at PS154 in Brooklyn – volunteer, 2022.

NJ Ocean Fun Days, Estuary Days, Environment Days – volunteer, 2017-19.

Young Women's Conference in Science, Technology, Engineering & Mathematics (Princeton Plasma Physics Laboratory) – volunteer, 2018-19.

Plainsboro Library – developed youth program “Motion in the Ocean”, 2017.

Future City – member of local nonprofit organization aimed at educating communities about environmental issues, working with policy-makers, and developing environmental initiatives, 2016-18.

Environmental Protection Agency: Trash Free Waters – attended meetings to discuss pollution issues facing New York and New Jersey waterways, 2017.